

scripts/ci.sh — User Guide

Last verified: 2026-05-16 (Phase 6a + 6b §11.4.29 snake_case migration: scripts/ci.sh path-references updated from Submodules/ to submodules/) **Inheritance:** HelixConstitution §11.4.18 (script documentation mandate)

Overview

Stub doc generated from the script's in-source header comment. See scripts/ci.sh for canonical behavior. This stub exists so the CM-SCRIPT-DOCS-SYNC pre-push gate (.githubhooks/pre-push Check 9) starts gating modifications to this script.

The script's own header documentation (verbatim):

```
scripts/ci.sh – local-only CI gate for Lava.
```

Per the Local-Only CI/CD constitutional rule, this script IS the project's CI/CD apparatus. The same script runs in three modes:

- | | |
|-----------------------------|---|
| <code>--changed-only</code> | Fast subset for the pre-push hook (Spotless, unit tests of changed modules, constitutional doc parser, forbidden-files check). No real-device tests; no mutation tests. |
| <code>--full</code> | All gates – unit tests across every module, parity gate, mutation tests where wired, fixture freshness, Compose UI Challenge Tests (requires a connected Android device or emulator). Used at tag time. |
| (default) | Same as <code>--full</code> . |

Per Sixth Law clause 5: passing CI is necessary, NOT sufficient for a release. The operator real-device verification per Task 5.22 of SP-3a is the load-bearing acceptance gate; this script certifies the codebase is shippable, not that the user-visible feature is shipped.

Usage

See the script's in-source comment block (above) for canonical usage examples.

Pipeline steps (current order)

1. Hosted-CI forbidden-files check (Local-Only CI/CD rule)
2. Host-power forbidden-command regex check (Host Stability rule)
3. Spotless (./gradlew spotlessCheck)
4. Unit tests across changed modules
5. Constitutional doc parser (scripts/check-constitution.sh)
6. **5a1 (added 2026-05-14):** §6.AC non-fatal-coverage scan (STRICT default; closes §6.AC-debt + CM-NONFATAL-COVERAGE gate)
7. **5a2 (added 2026-05-15):** §6.AB Challenge-discrimination scan (STRICT default; closes §6.AB-debt + CM-CHALLENGE-DISCRIMINATION gate)
8. **5a3 (added 2026-05-15, 31st §6.L):** §6.AE per-feature Challenge coverage scan (STRICT default after per-feature backfill drained the queue 2026-05-15; CM-CHALLENGE-COVERAGE gate; LAVA_CHALLENGE_COVERAGE_STRICT=0 to revert to advisory)
9. **5a4 (added 2026-05-15, constitution-compliance plan Phase 1):** §11.4.32 verify-all-constitution-rules sweep (STRICT default; the §11.4.32 mandated enforcement engine for every other constitution rule; wraps every individual gate above + every hermetic test suite into one invocation; produces per-run attestation JSON at .lava-ci-evidence/verify-all/<UTC-timestamp>.json; CM-VERIFY-ALL-CONSTITUTION-RULES gate)
10. 5b. Hermetic bash test suites (under tests/)
11. (Full-mode only) Parity, mutation, fixture-freshness, Compose UI Challenge Tests on connected device

Forbidden-files check scoped to git ls-files — added 2026-08-20

Fixes a real false-positive in Pipeline step 1: the check previously used a raw `find .` over the working directory, which also walked (a) other worktrees checked out under `.claude/worktrees/agent-*/` — each is a DIFFERENT branch's file tree, physically nested under this repo's working directory by the harness but irrelevant to what the CURRENT branch is about to push — and (b) vendored third-party content inside nested submodules-of-submodules (e.g. HelixQA's `tools/opensource/*`, which legitimately ships upstream `.github/workflows/*` files as that nested repo's own concern). Both classes produced spurious FORBIDDEN HOSTED-CI FILES rejections on a push whose own tracked tree contained zero

such files. Fixed by scoping the scan to `git ls-files` (this branch's actually-tracked files only) — submodules naturally excluded as gitlinks (not descended into without `--recurse-submodules`, which is deliberately not passed; submodule-internal governance is scoped separately per §6.F), other worktrees naturally excluded since their files aren't part of this branch's tree at all.

Bluff-Audit: `scripts/ci.sh` Hosted-CI forbidden-files check Mutation: `git add -f a real .github/workflows/mutation-test.yml` fixture into the tracked tree, confirming the NEW `git ls-files`-scoped check still correctly DETECTS a genuine violation (true-positive preserved). Observed-Failure: N/A for this direction (positive detection confirmed); the OLD `find .`-based check was separately confirmed to false-positive on real, pre-existing, untracked content under `.claude/worktrees/` and `submodules/helixqa/tools/opensource/` during an actual `git push` attempt against this exact commit range — that rejection is the falsifiability evidence for the bug this fix closes. Reverted: yes — the mutation fixture was `git reset + rm -rf`'d immediately after the positive-detection check passed; confirmed via `git status --short` showing a clean tree before this commit.

§6.AE matrix gate (separate entry point)

For §6.AE.2 gate-mode runs producing per-AVD attestation: `bash scripts/run-challenge-matrix.sh`. The runner correctly REFUSES to claim success on hosts lacking KVM (darwin/arm64) — exits 2 with a host-gap diagnostic. Real attestations require a Linux `x86_64 + KVM` gate-host.

Maintenance

When this script is modified, update this document in the same commit (CM-SCRIPT-DOCS-SYNC requires it). Per §11.4.18, the documentation MUST stay in sync with the codebase — no doc may be out of sync with its script.

Cross-references

- `scripts/ci.sh` — the script itself
- `docs/helix-constitution-gates.md` — gate inventory
- HelixConstitution Constitution.md §11.4.18 (the mandate)
- Lava CLAUDE.md §6.AD (HelixConstitution Inheritance)